

C# LINQ COMPLETE CHEAT SHEET

USING:

```
using System.Linq;
```

1. FILTERING

Where()

```
var result = numbers.Where(x => x > 5);
```

2. PROJECTION

Select()

```
var squares = numbers.Select(x => x * x);
```

SelectMany()

```
var allSubjects = students.SelectMany(s => s.Subjects);
```

3. SORTING

OrderBy()

OrderByDescending()

ThenBy()

4. ELEMENT OPERATORS

First()

FirstOrDefault()

Single()

SingleOrDefault()

Last()

ElementAt()

5. QUANTIFIERS

Any()

All()

6. AGGREGATES

Count()

Sum()

Min()

Max()

Average()

Aggregate()

7. GROUPING

GroupBy()

8. JOINING

Join()
GroupJoin()

9. SET OPERATORS

Distinct()
Union()
Intersect()
Except()

10. PARTITIONING

Take()
Skip()
TakeWhile()
SkipWhile()

11. CONVERSION

ToList()
ToArray()
ToDictionary()
ToLookup()

12. CONCAT & ZIP

Concat()
Zip()

13. EXECUTION TYPES

Deferred Execution
Immediate Execution (ToList, ToArray)

IMPORTANT CONCEPTS

- LINQ uses Generics
- Uses Extension Methods
- Uses Lambda Expressions
- Works on IEnumerable<T>
- IEnumerable = Deferred execution
- IQueryable = Database querying

INTERVIEW SUMMARY:

LINQ provides a unified, type-safe way to query data from collections, XML, and databases.